Physics-Informed Machine Learning Framework for Predicting Micro-viscoelastic and Micro-structural Properties of Granular Composites

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Abstract: This study develops a Physics-Informed Machine Learning (PIML) framework for the quantitative characterization and real time quality monitoring of micro-viscoelastic granular materials, with emphasis on compressed oral solid dosage (OSD) forms (tablets). The ultrasonic attenuation and dispersion responses of the OSD forms are interpreted through time-domain and frequency-domain approaches. By employing spectral-domain waveform inversion, key mechanical parameters including bulk Young's modulus, relaxation times, and grain-size distribution are extracted from experimental measurements. In this work, two Machine Learning (ML) techniques based on the extracted micro-viscoelastic, scattering properties and dispersion model, are demonstrated. Synthetic waveforms with a given set of properties of virtual tablets are computationally generated to train, validate, and test the developed ML models for the granular composites in reconstructing original waveform. The resulting robust PIML framework enables real-time, non-destructive ultrasonic evaluation (NDE) of granular composite materials, providing a powerful tool for in-line quality monitoring and process optimization in pharmaceutical manufacturing of OSD forms.

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